Lec 15, 16: Global Nonlinear stability.

Liapunov stability

Ref: See 9.2. [Smale].

see also. Sec 9.6. [BDM]

Liapunov stability: Consider

(1) $$\frac{d}{dt}X(t) = F(x)$$

- Previous stability result: require $|X(0) - X_*|$ small.

- Consider C^1 function. $L: O \to R$ (open set, scalar) and $L(X(t))$

e.g. $L(Y) = \sum Y_i^2$,

- $\frac{d}{dt}L(X(t)) = \dot{X}(t) \cdot (\nabla L)(X(t)) = \vec{F}(X(t)) \cdot (\nabla L)(X(t))$.

(2) Denote $\dot{L}(x) = \vec{F}(x) \cdot (\nabla L)(x)$.

Thm: (Liapunov stability).

Let x^* be equilibrium. Suppose $L: O \to R$ differentiable.

(Ass 1). $L(x^*) = 0$, $L(x) > 0$, for $x \neq x^*$

(Ass 2): $\dot{L} \leq 0$ in $O \setminus \{x^*\}$.

Then x^* is stable.

(Ass 2)': $\dot{L} < 0$ in $O \setminus \{x^*\}$

Then x^* is asymptotically stable.

- Idea: $\overset{\text{Ass 1}}{(1)}$ $L(x)$ measure size of $x - x^*$, e.g. $L(x) = \sum (x_i - x_{*,i})^2$ satisfies (1)

(2) $L(x(t))$ is monotone decreasing along solution curve.

Application.

Pendulum:

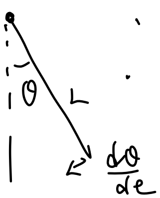
$$m \frac{d^2\theta}{dt^2} + \frac{mg}{L} \sin\theta = 0.$$

$$\vec{F}(x,y) = \begin{pmatrix} y \\ -\frac{g}{L}\sin x \end{pmatrix}$$

- $x = \theta$, $y = \frac{d\theta}{dt}$. $\Rightarrow \begin{cases} \frac{dx}{dt} \\ \frac{dy}{dt} \end{cases} = \begin{matrix} y \\ -\frac{g}{L}\sin x \end{matrix}$

- Potential energy due to gravity

$$U(x,y) = mgL(1-\cos x) \geq 0.$$



- U is smallest when $x = \theta = 0$
- $x = \theta = -\pi$ Largest potential

- kinetic energy $= \frac{1}{2} m v^2$. $v = L\frac{d\theta}{dt}$.

Total energy. $V = U(x,y) + \frac{1}{2} m L^2 y^2$

$$= mgL(1-\cos x) + \frac{1}{2} m L^2 y^2$$

- consider equil. $x = 0$, $y = 0$

$$O = (-\pi/2, \pi/2) \times \mathbb{R}$$

Ass 1: $V > 0$: $O \setminus \{0,0\}$. $V(0,0) = 0$ ✓

Ass2: Check: $\vec{F} = \begin{pmatrix} y \\ -\frac{g}{L}\sin x \end{pmatrix}$

$\partial_x V = mgL\sin x$

$\partial_y V = mL^2 y$.

$\vec{F} \cdot \nabla L = 0 \Rightarrow$ Ass 2 is satisfied

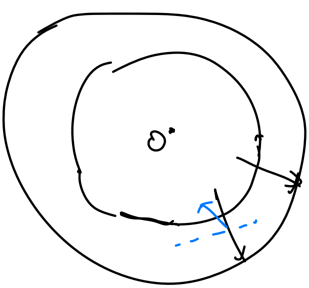
Yield. $V(x(t), y(t)) = V(x(0), y(0))$!

Thus stable.

- For $x = 2n\pi + \pi$, show unstable.

$$\frac{d}{dt}\vec{x} = \vec{F}(\vec{x}).$$

Geometry meaning: $\frac{d}{dt} L = \vec{F} \cdot \nabla L \leq 0$



→ Level set of L

∇L direction

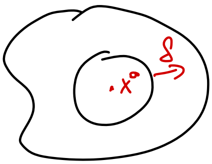
direction of $\vec{F}$. pointing inward.

pf of Thm:

Step 1: Compute

$$\frac{d}{dt} L(x(t)) = (\vec{F} \cdot \nabla L)(x(t)) = \dot{L}(x(t))$$

Step 2: Proof of stable. Fix small $\delta > 0$ s.t.



$B_{x^*}(\delta) \subseteq O$

. Denote $\alpha = \min_{x \in \partial B_0(\delta)} L(x) > 0.$

Now want to prove if $|X_0 - X^*| < \varepsilon \Rightarrow |X(t) - X^*| < \delta$

Step 3

(Ass 1) + Continuity $\Rightarrow$ Initial point.

$|L(x_0)| \leq |L(x_0) - L(x^*)| < \alpha \quad \forall |x_0 - x^*| < \varepsilon$

$B_\delta(x^*) \subseteq O$

Since

(Ass 2) $\Rightarrow \frac{d}{dt} L(x(t)) \leq 0$

$\Rightarrow L(x(t)) \leq L(x_0) < \alpha, \quad |x_0 - x_*| < \varepsilon$

$\Rightarrow X(t)$ can not leave $B_\delta(x^*)$ (Why)

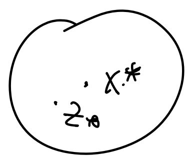
$\Rightarrow |X(t) - x_*| < \delta \quad , \forall t \geq t_0$

Step 4: If not hold $x(t) \longrightarrow x_*$. $\quad X(t) \in B_{x^*}(\delta)$

Then $\exists\, t_n \longrightarrow \infty$ s.t

$X(t_n) \longrightarrow z_* \neq x_*$

$\in B_{x_*}(\delta)$



. Note let $z(s)$ be solution to

$\dot{z} = F(t, z(t)), \quad z(0) = z_*$

Then $L(z(s)) < L(z(0)) = L(z_*) \quad \forall s > 0$

- Compare 2 solu. $X(t_n+s)$, $Z(s)$

since $X(t_n) - Z(0) = X(t_n) - Z_*$ arbitrary close

- L: Lip $\Rightarrow$

$$L(X(t_n+1)) - L(Z_*) \quad \overset{\Delta}{=} -a < 0$$

$$= L(X(t_n+1)) - L(Z(1)) + L(Z(1)) - L(Z_*)$$

$$\leq C \cdot |X(t_n+1) - Z(1)| - a$$

$$\leq C \cdot |X(t_n) - Z_*| - a < 0 \quad \text{as } n \to +\infty$$

contradict: $L(X(t_1)) \geq L(Z_*)$

Drop a little.

Example: Do many Examples.

$$x' = (\varepsilon x + 2y)(z+1)$$
$$y' = (-x + \varepsilon y)(z+1)$$
$$z' = -z^3 \qquad . \ \varepsilon \text{ parameter.}$$

- $(0,0,0)$ equilibrium
- Linearization:

$$Y' = \begin{pmatrix} \varepsilon & 2 & 0 \\ -1 & \varepsilon & 0 \\ 0 & 0 & 0 \end{pmatrix} Y.$$

- unstable $\varepsilon > 0$
- Find Liapnov func.

$$L(x,y,z) = ax^2 + by^2 + cz^2 \text{ form}$$

- $$\frac{\dot{L}}{2} = \varepsilon(ax^2 + by^2)(z+1) + (2a-b)xy(z+1) - cz^4.$$

$2a - b = 0$. Start $z+1 > 0$, $\Rightarrow \dot{L} < 0$.

- Asymptotic stable for (x,y,z) with $z > -1$.

- Ex 2:

$$x' = -x^3 \qquad \text{Do} \quad x^2 + y^2 + z^2$$
$$y' = -y(x^2 + z^2 + 1)$$
$$z' = -\sin z.$$

- Ex3: In many cases, need

$$V = ax^2 + bxy + cy^2 \geq 0. \quad (4)$$

- $V > 0 \quad \forall x, y \iff a, c > 0, \quad 4ac > b^2.$

complete square.

Asymptotic stable.

Ex 4.

- $\frac{dx}{dt} = -x - xy^2, \qquad \frac{dy}{dt} = -y - x^2 y$

- Method 1: Small region Jacobian $\nabla F|_{(0,0)} = \begin{pmatrix} -1 & \\ & -1 \end{pmatrix}$
- Method 2: Lyapunov:

  Now try to find a, b, c in (4) s.t.

Compute $\dot{V} = \partial_x V \cdot \dot{x} + \partial_y V \cdot \dot{y} = \nabla V \cdot \vec{F}$

$V = x^2 + y^2$

Ex5. $\frac{dx}{dx} = -x - xy^2, \qquad \frac{dy}{dt} = -y - x^2 y + \mu \cancel{x}.$

any $\mu > 0$

Try: $V = ax^2 + y^2 \quad a$ large.